Insurance Portfolio Performance Dashboard

IslandGuard Insurance Ltd — Mauritius — 2015 to 2025

Power BI | Power Query | DAX | SQL | Star Schema Modelling

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Programme	MSc Data Analytics
Domain	Finance, Insurance & Reinsurance
Dataset	10,000 policies | 9,945 customers | MUR currency | 11 years (2015–2025)
Dashboard	3 pages — Executive Summary | Insurance Overview | Risk & Retention
Objective	End-to-end BI solution enabling IslandGuard management to monitor premium income, risk concentration, and customer retention

Contents

1. Project Context

This project demonstrates a complete, production-grade Business Intelligence workflow — from blank dataset through to executive-ready dashboard — applied to a domain that directly mirrors the analytical challenges faced by employers in banking, insurance, and financial services.

The dataset was built from scratch. Starting from an empty insurance policy template, every customer record, premium value, district assignment, and agent profile was authored to reflect realistic Mauritian market conditions — 10,000 policies, 9,945 customers, 22 named agents, MUR currency, and 12 authentic Mauritius districts across an 11-year period (2015–2025). The result is a fully localised, analytically rich dataset structured as a seven-table star schema.

The insurance domain was chosen because its analytical framework — risk segmentation, premium analysis, retention modelling, and exposure quantification — transfers directly across four target employer sectors:

Sector	How This Project Applies
Banking	Bancassurance premium analysis, customer retention, portfolio risk segmentation
Finance	Investment-linked policy ROI, CAGR analysis, portfolio performance tracking
Insurance	Premium income, surrender/lapse rates, underwriting expense, agent performance
Reinsurance	Aggregate risk exposure, maturity liability, geographic risk concentration

> ***Core business question: "How is IslandGuard's insurance portfolio performing across premium income, risk exposure, and customer retention — and where does management need to act?"***

2. Stakeholder Requirements

The project opened with a structured requirements-gathering exercise, mapping the information needs of four distinct stakeholder personas at IslandGuard. Each persona drove specific dashboard design decisions — from which KPI cards appeared on the Executive Summary to which drill-down dimensions were made available on the Risk & Retention page.

Stakeholder	Primary Questions
Chief Financial Officer	What is our total annual premium income? What are our underwriting expenses? What is our forward premium payable pipeline?
Head of Risk & Actuarial	What is our total maturity liability (Sum Assured)? Where is risk geographically concentrated? What is our CAGR across the active portfolio?
Customer Retention Manager	What is our surrender rate? Which districts and products have the highest lapse risk? Which agents have the most at-risk policies?
Sales Director	Which agents are top performers by premium? Which policy types are selling best? How is premium trending year on year?

Three success criteria were agreed before any data work began: the dashboard had to consolidate all key KPIs into a single self-service tool; it had to allow non-technical stakeholders to filter and explore without analyst intervention; and it had to flag underperformance automatically through visual design rather than requiring interpretation.

3. Data Design & Preparation

3.1 Dataset Architecture

The data model is a classic star schema — one central fact table surrounded by six dimension tables, joined on business keys. This structure was chosen because it enables efficient DAX calculations, clean filter propagation, and a model that could connect to live data sources in a production environment with minimal rework.

Table	Type	Records	Purpose
FCT Insurance_Policy_Table	Fact	10,000	Core policy records — premiums, status, dates, tenure
DM Customer_Detail_Table	Dimension	9,945	Customer demographics, occupation, district
DM Insurance_Agent_Table	Dimension	22	Named Mauritian agents across all 12 districts
DM Policy_Protection_Plan	Dimension	8	IslandGuard product names and codes
DM Policy_Type	Dimension	3	Endowment, Whole Life, Universal Life
DM Regional_Manager	Dimension	5	North, South, East, West, Central regions
DM Zonal_Manager	Dimension	3	Zonal leadership structure

3.2 Data Issues Identified & Resolved

Five data quality issues were identified during profiling — each one diagnosed to root cause and resolved before any measures were built. This diagnostic process is documented because it reflects the real analytical work: the dashboard's accuracy depends entirely on decisions made here.

Issue Found	Business Impact	Resolution Applied
Date columns stored as text (DD-MM-YYYY)	DATEDIFF and time intelligence functions returned blanks across all date measures	Applied Date.FromText with explicit format specification in Power Query for all three date columns
Payment Frequency SWITCH missing Monthly case	All Monthly policies silently returned BLANK(), cascading nulls into	Added Monthly × 12 case; changed fallback from BLANK() to 0

	premium and ROI calculations	
Broken Agent–Policy relationship	Treemap showing equal premium values for all agents — relationship not established in model	Established correct one-to-many join on Agent Code in Model view
Non-Active policies in summary table	Blank rows appearing for Surrendered and Lapsed policies in executive visuals	Applied report-level filter: Policy Status = Active, with page-level override on Risk & Retention
Duplicate Merge1 query in Power Query	Risk of duplicate rows and model confusion from accidental merge operation	Identified and deleted from Power Query Editor before it could affect calculations

4. DAX Measures & Business Logic

Twelve measures were developed to answer the four stakeholder question sets. All Active-policy measures use CALCULATE or FILTER to enforce business rules: no loan data, no claims data, and premiums annualised by payment frequency (Monthly × 12, Quarterly × 4, Annually × 1). The discipline of writing every complex measure using VAR blocks — decomposing each calculation into named, single-purpose variables — made debugging and peer review significantly faster.

Measure	Business Meaning	Key DAX Functions
Total Annual Premium	All active premiums normalised to annual value	SUMX, SWITCH, FILTER
Total Premium Amount	Full premium commitment over policy lifetime	SUMX, VAR
Total Premium Paid	Actual collections from policy start to today — capped at tenure	SUMX, DATEDIFF, MIN, VAR
Premium Amount Payable	Remaining premiums owed, floored at zero	SUMX, MAX, VAR, DATEDIFF
Premium Payment Duration	Average years policyholders have been actively paying	AVERAGEX, DATEDIFF, MIN
Total Maturity Amount (MUR)	Total Sum Assured across active policies — the full liability figure	CALCULATE, SUM
Annualised ROI %	Annual return per policy relative to premiums paid	AVERAGEX, DIVIDE, VAR
CAGR %	Compound annual growth rate from premium paid to sum assured	AVERAGEX, POWER, DIVIDE, IF
Surrender Rate %	Surrendered policies as % of total book	CALCULATE, COUNTROWS, DIVIDE
Lapse Rate %	Lapsed policies as % of total book	CALCULATE, COUNTROWS, DIVIDE

Policies at Risk	Combined count of Surrendered and Lapsed policies	CALCULATE, COUNTROWS
Premium Lost (MUR)	Total annual premium from at-risk policies	CALCULATE, SUMX

The most technically demanding measure was CAGR, implemented as POWER(EndValue/StartValue, 1/Tenure) − 1, wrapped in an IF guard against zero-tenure edge cases. SUMX was used throughout in preference to SUM wherever row-level iteration was required. DIVIDE replaced the division operator on every rate calculation to handle zero denominators gracefully.

5. Dashboard Design

The three-page structure maps directly to the three stakeholder tiers identified in Section 2 — executive, analytical, and operational. Each page has a single design objective and a defined primary audience.

Page 1 — Executive Summary

Built for the CFO and senior leadership. Six KPI cards across the top row deliver an immediate portfolio snapshot: Policies (7.33K), Total Premium (7.78bn MUR), Annual Premium (524M MUR), Premium Paid (2.59bn MUR), Premium Payable (16.34bn MUR), and Underwriting Expense (35.52M MUR). Six interconnected slicers — Policy Type, Policy Name, Sales Agent, Year, District, Occupation — enable full self-service filtering without requiring analyst support. A detailed policy-level table beneath the slicers gives operational users access to calculated values per customer.

Page 2 — Insurance Overview

Designed for analysts and the Sales Director requiring dimensional breakdowns and trend context. Donut charts surface product-type mix and individual product performance. A relationship-joined treemap shows agent premium contribution once the Agent dimension was correctly connected to the fact table — a fix that transformed an uninformative equal-value chart into one of the most actionable visuals in the dashboard. Bar charts break premium down by district (MUR millions) and customer occupation. A line chart tracks annual premium from 2015 to 2024, with 2025 excluded as a partial year.

Page 3 — Risk & Retention

Built for the Head of Risk and Customer Retention Manager. Four KPI cards lead: Policies at Risk (3K), Lapse Rate (1.15%), Surrender Rate (25.21%), and Premium Lost (637.90M MUR). Five drill-down visuals expose the where, which, and who of portfolio attrition: surrender rate by district, at-risk policies by product, surrender rate by policy tenure, at-risk policy trend 2015–2024, and at-risk policies by agent. A page-level filter override allows Surrendered and Lapsed policies to appear here despite the report-level Active filter applied across pages one and two.

6. Key Findings & Recommendations

The dashboard exposed a business in good premium health but with a retention crisis that, left unaddressed, will compound into a structural revenue problem. The findings below are ordered by financial materiality.

⚠ Critical Finding

Surrender rate of 25.21% — more than double the 8–12% industry benchmark — has resulted in MUR 637.9 million in lost annual premium across 3,000 policies. At current trajectory, this is not an anomaly or a product design issue. The surrender rate is consistent across all policy tenure bands (24–27% regardless of commitment length), which points to a systemic cause: service quality, competitor pressure, or pricing misalignment — not policy structure.

Finding	Business Impact	Recommended Action
Surrender rate of 25.21% — more than double industry benchmark of 8–12%	MUR 637.9M in lost annual premium; 3,000 active policies at attrition risk	Proactive retention programme — outreach to at-risk policyholders 90 days before expected lapse
Surrender rate consistent across all tenure bands (24–27%)	Systemic issue — not product design or commitment length — likely service quality or competitor pressure	Conduct customer exit interview analysis; benchmark service quality and pricing against direct competitors
Quatre Bornes (28.9%), Plaines Wilhems (26.2%) and Flacq (26.1%) lead surrender by district	Urban districts with higher-income customers have more financial product alternatives — accelerating churn	Deploy targeted retention campaigns in these three districts with dedicated relationship manager assignment
Income Protection (361 surrenders) and ULIP Growth (349) are the most surrendered products	Disproportionate revenue leakage relative to policy count — product competitiveness at risk	Review pricing and benefit structure against market; consider premium tier restructuring
Premium growth plateaued from 2022 — peaked at 183M MUR, now stable around 175M MUR	Potential market saturation in core districts; growth ceiling approaching	Develop geographic expansion into underserved districts: Savanne, Black River, Rivière du Rempart
Whole Life policies account for 37.3% of annual premium — single product concentration	Significant exposure if market shifts toward investment-linked products	Drive ULIP and Income Protection through agent incentive structures to diversify product mix
Top agents each hold 130+ at-risk policies — no retention KPIs in place	Agent incentives currently reward acquisition, not retention — misaligned with business health	Introduce retention KPIs into agent performance framework alongside premium volume targets

7. Skills Demonstrated

Business Analysis	Technical
• Requirements definition and stakeholder persona mapping • Dataset design and population from blank templates — full data ownership • Data profiling, root cause diagnosis, issue resolution • Multi-dimensional financial insight generation • Business recommendations with quantified financial impact (MUR) • Stakeholder-oriented design across three audience types	• Power BI Desktop — data model, visuals, slicers, conditional formatting • Power Query (M) date parsing, type conversion, query cleanup • DAX — SUMX, CALCULATE, AVERAGEX, VAR, POWER, DIVIDE, DATEDIFF • Star schema modelling — 1 fact table, 6-dimension tables, 6 relationships • Filter context management — visual, page, and report level • GitHub version control and project documentation